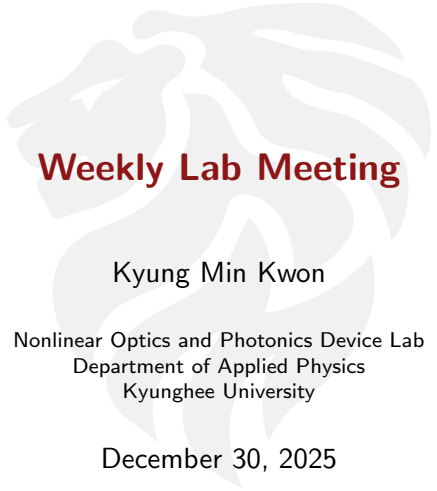




Weekly Lab Meeting

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December 30, 2025



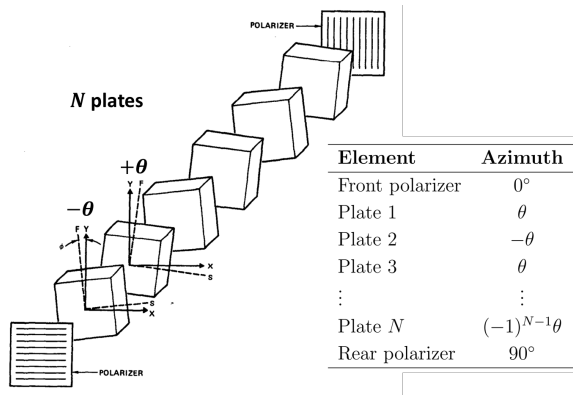
Review on Šolc Filter

Šolc Filter

• Šolc Filter

Composed of a pile of identical birefringent plates each oriented at a prescribed azimuth angle θ . (θ : rocking angle)

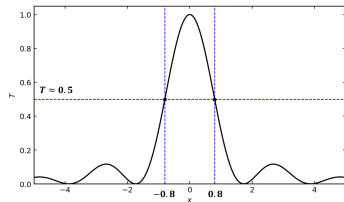
Two types: **folded type** and fan type.



Analysis on Šolc Filter

- Spectral Filter: FWHM ($x = N\Delta\Gamma/\pi$)

$$T = \left| \frac{\sin\left(\frac{1}{2}\pi\sqrt{1+x^2}\right)}{1+x^2} \right|^2 \rightarrow T = \sin^2 2N\theta$$



Fundamental wavelength under the HWP setting.
(d : design parameter for the target wavelength.)

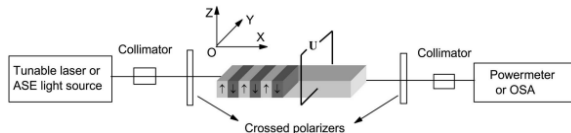
$$\lambda_0 = \frac{2}{2m+1}(n_o - n_e)d \quad (1)$$

Full Width Half Maximum (FWHM)

$$\Delta\lambda = 1.60 \frac{\lambda_\nu}{(2\nu+1)N} \quad (2)$$

PPLN EO Šolc Filter

- (X. Chen et al., 2003)



Propagating in the x -direction.

The presence of an external electric field along the Y axis.

→ Y and Z optical principal axes are slightly rotated about the X -axis.

$$\theta \approx \frac{r_{51} E}{(1/n_e^2) - (1/n_o^2)} \quad (3)$$

(1) proportional to the intensity of the applied voltage

(2) Very narrow spectrum filters can be achieved by longer PPLN crystal (N, ν)

PPLN EO Šolc Filter

- Derivation of EO induced rocking angle

$$r_{mk}E_k = \begin{pmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \\ 0 & r_{42} & 0 \\ r_{42} & 0 & 0 \\ r_{22} & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ E \\ 0 \end{pmatrix} = \begin{pmatrix} -r_{22}E \\ r_{22}E \\ 0 \\ r_{42}E \\ 0 \\ 0 \end{pmatrix} \quad (4)$$

Impermeability tensor

$$\eta(\mathbf{E}) = \begin{pmatrix} 1/n_o^2 - r_{22}E & 0 & 0 \\ 0 & 1/n_o^2 + r_{22}E & r_{42}E \\ 0 & r_{42}E & 1/n_e^2 \end{pmatrix} \quad (5)$$

PPLN EO Šolc Filter

- Derivation of EO induced rocking angle

Transverse impermeability tensor

$$\eta(\mathbf{E}) \begin{pmatrix} 1/n_o^2 + r_{22}E & r_{42}E \\ r_{42}E & 1/n_e^2 \end{pmatrix} \quad (6)$$

The rocking angle is defined as:

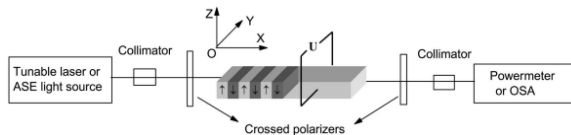
$$\tan 2\theta = \frac{2\eta_{t,12}}{\eta_{t,11} - \eta_{t,22}} = \frac{2r_{42}E}{(1/n_o^2) - (1/n_e^2) + r_{22}E} \approx \frac{2r_{42}E}{(1/n_o^2) - (1/n_e^2)}$$

For small angle θ ,

$$\theta \approx \frac{r_{42}E_1}{(1/n_1^2) - (1/n_3^2)} \quad (7)$$

Experimental Setup of PPLN EO Šolc Filter

- Experimental setup



The device has dimensions of $28 \text{ mm} \times 5 \text{ mm} \times 0.5 \text{ mm}$ and consists of four gratings with periods of $20.2 \mu\text{m}$, $20.4 \mu\text{m}$, $20.6 \mu\text{m}$ and $20.8 \mu\text{m}$.

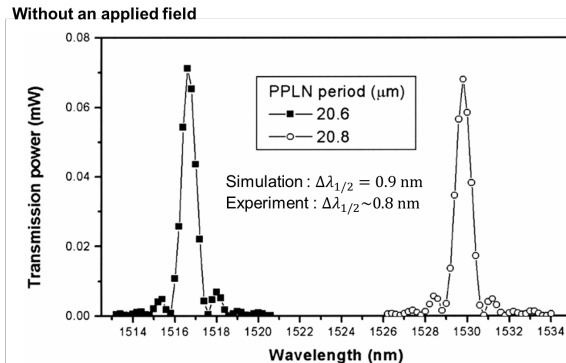
20.6 to 20.8 mm and a width of 1 mm (1525.5 and 1511.2 nm)

Front (end) polarizer : Y (Z) axis.

Wavelength values at 24°C are calculated with the Sellmeier equation.

Experimental Result of PPLN EO Šolc Filter

• Experimental Result : Without an Applied Field



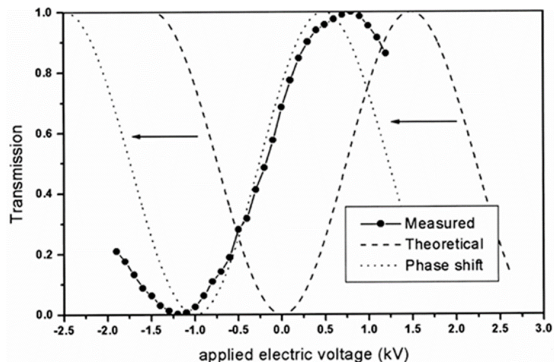
The transmission of the power is measured to be 70 %.

The origin of the initial rocking angle θ is not clear.

→ Photorefractive effect (Chen et al., 2006)

Experimental Result of PPLN EO Šolc Filter

- Experimental Result : Amplitude Modulator



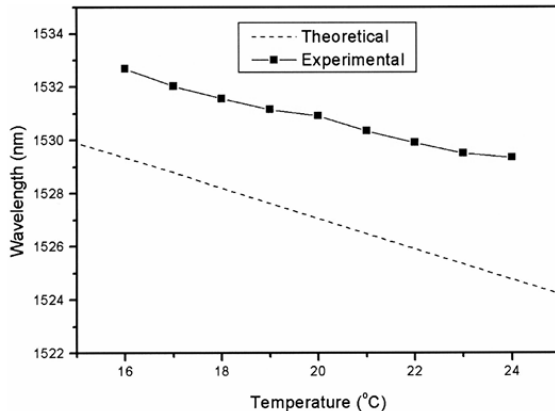
28 μm PPLN @ 24°C

Phase shift : 0.009 rad

Filter is blocked @ -1.2kV and 100% transmission @ 0.75kV

Experimental Result of PPLN EO Šolc Filter

- Experimental Result : Temperature Dependence of $28\text{ }\mu\text{m}$ PPLN



The ordinary and extraordinary indices of LN are temperature dependent.

→ Central wavelength is dependent on temperature.

Comment of PPLN EO Šolc Filter

- **Initial rocking angle**

The photovoltaic effect (PVE) plays an essential role in the performance of a **non-applied field**, which is substantiated by subsequent experiments.

Phase shift is observed. The PVE is attributed this phase difference (0.991 rad).

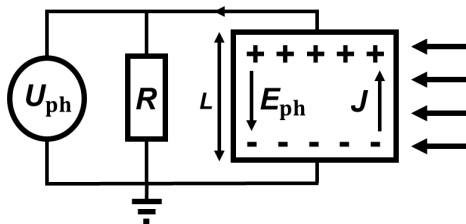
What is the **Photovoltaic Effect**?

A stylized, light gray lion head logo is centered in the background. The lion is facing left, with its mane flowing back. The text "Photovoltaic Effect" is overlaid on the center of the lion's face.

Photovoltaic Effect

Photovoltaic Effect (Fridkin, 2001)

- Photovoltaic Effect



Illumination of the crystal generates charge carriers (electrons and holes).

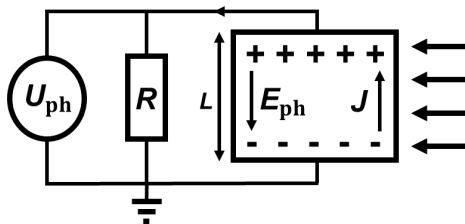
Short circuit ($R = 0$) : records the photocurrent J .

Open circuit ($R = \infty$) : the photovoltage (or the corresponding field) U_{ph} (E_{ph}).

→ **Photovoltaic effect**

Photovoltaic Effect (Fridkin, 2001)

- Photovoltaic Effect



Dember effect : diffusion of nonequilibrium carriers and the variations in the mobility of holes and electrons.

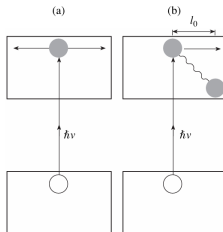
Nonequilibrium carriers at crystal inhomogeneities.

→ These are not in our case.

Bulk photovoltaic effect: It occurs in noncentrosymmetric crystals and can be observed even in a homogeneous crystal under uniform illumination.

Bulk Photovoltaic Effect

- A qualitative description



The violation of the principle of the detailed balancing for photoexcited carriers

The probability $W_{k,k'}$ of the electron transition from the state with a momentum k' to a state with the momentum k .

Centrosymmetry : The energy band is symmetric ($W_{k,k'} = W_{k',k}$), No current by $\langle \mathbf{v} \rangle = 0$.

Non-centrosymmetry : Asymmetric momentum distribution ($W_{k,k'} \neq W_{k',k}$) with $\langle \mathbf{v} \rangle \rightarrow$
Relaxed electrons go to the band bottom $\rightarrow$ shift $l_0 \rightarrow$ photocurrent (photovoltage)

Linear Equation on Bulk Photovoltaic Effect

- The bulk photovoltaic effect with linear polarized light

$$J_i = \alpha G_{ijl}^L e_j e_l^* I + i \alpha G_{ijl}^C [\mathbf{e} \mathbf{e}^*]_l I = \alpha G_{ijl} E_j E_l^* \quad (8)$$

The allowance for the quadratic dependence of the current J on the field due to the light wave with the unit vector ($\mathbf{e}$) for linearly (L) and circularly (C) polarized light.

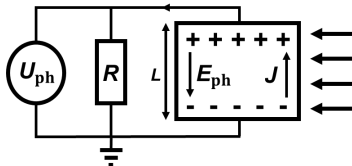
The linear bulk photovoltaic effect is nonzero only in noncentrosymmetric crystals.

G_{ijl} is the conventional piezoelectric tensor.

Photovoltage by Bulk Photovoltaic Effect

- The electrodes are disconnected (without an applied field)

After enough times, the current will lead to the accumulation of charges at the edge of the illuminated area.



$$U_{ph} = J_i L / \sigma \quad (9)$$

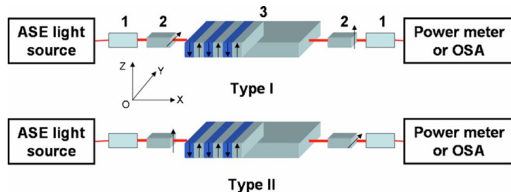
$$E_{ph} = -J_i / \sigma = -\alpha G_{jl} E_j E_l^* / \sigma \quad (10)$$

where dark conductivity and photoconductivity $\sigma = \sigma_d + I\sigma_{ph}$.

Under illumination with open circuit condition, the bulk photovoltaic effect (BPVE) can generate a space-charge field even without an applied field.

PPLN PVE Šolc Filter (Chen et al., 2006)

- Linear equation of PVE



$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \alpha \begin{pmatrix} 0 & 0 & 0 & 0 & G_{15} & -G_{22} \\ -G_{22} & G_{22} & 0 & G_{15} & 0 & 0 \\ G_{31} & G_{31} & G_{33} & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} E_1 E_1^* \\ E_2 E_2^* \\ E_3 E_3^* \\ E_1 E_2^* \\ E_1 E_3^* \\ E_2 E_3^* \end{pmatrix}. \quad (11)$$

PPLN PVE Šolc Filter

- **Case study**

For type 1 (y to z-polarized light),

$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ G_{22} \\ G_{31} \end{pmatrix} E_2 E_2^* \rightarrow E_{\text{ph}} = -\alpha G_{22} E_2 E_2^* \quad (12)$$

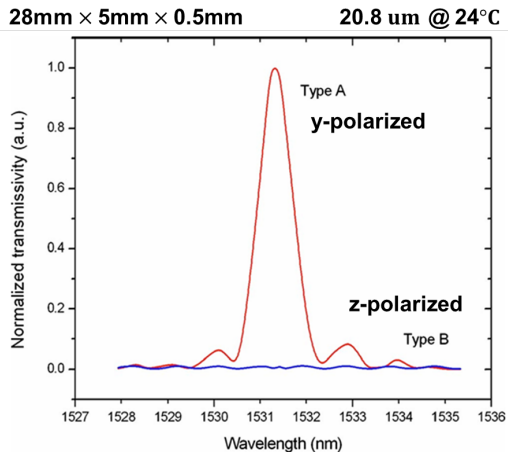
For type 2 (z to y-polarized light),

$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ 0 \\ G_{33} \end{pmatrix} E_3 E_3^* \rightarrow E_{\text{ph}} = 0 \quad (13)$$

For z-polarized light, no field will cause the half-wave plate rocking.

PPLN PVE Šolc Filter

- Result



In addition, there may be fabrication imperfections.



Simulation on Periodically Poled Šolc Filter

Plan for PPKN Simulation

• Plan

- (1) The numerical result of the photovoltaic effect (PVE) is unknown. So first, we first verify whether the PVE is insensitive to the input polarization.
- (2) Perform simulations for a 180° -domain PPKN device at 24°C and compare the results with those of PPLN.
- (3) Simulate the temperature dependence of the 180° -domain PPKN device.
- (4) Repeat them for the 90° -domain PPKN device.

Plan for PPKN Simulation

- Photovoltaic tensor of PPKN (mm2)

$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \alpha \begin{pmatrix} 0 & 0 & 0 & 0 & G_{15} & 0 \\ 0 & 0 & 0 & G_{24} & 0 & 0 \\ G_{31} & G_{32} & G_{33} & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} E_1 E_1^* \\ E_2 E_2^* \\ E_3 E_3^* \\ E_2 E_3^* \\ E_1 E_3^* \\ E_1 E_2^* \end{pmatrix}. \quad (14)$$

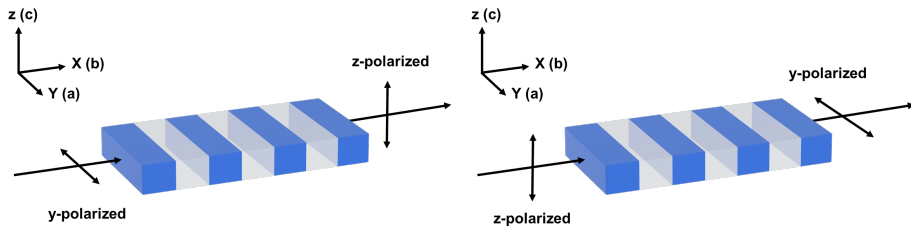
For y and z-polarized light in the x-direction,

$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ 0 \\ G_{32} \end{pmatrix} E_2 E_2^* \rightarrow E_{\text{ph}} = 0 \quad (15)$$

$$\begin{pmatrix} J_1 \\ J_2 \\ J_3 \end{pmatrix} = \alpha \begin{pmatrix} 0 \\ 0 \\ G_{33} \end{pmatrix} E_3 E_3^* \rightarrow E_{\text{ph}} = 0 \quad (16)$$

Plan for PPKN Simulation

- Photovoltaic tensor of PPKN (mm²)



Both y-polarized and z-polarized input states are free from PVE.

Plan for PPKN Simulation

- Recall Pockels tensors of mm2

for +c domain of PPKN,

$$r_{mk}E_k = \begin{pmatrix} 0 & 0 & r_{13} \\ 0 & 0 & r_{23} \\ 0 & 0 & r_{33} \\ 0 & r_{42} & 0 \\ r_{51} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} = \begin{pmatrix} r_{13}E_3 \\ r_{23}E_3 \\ r_{33}E_3 \\ r_{42}E_2 \\ r_{51}E_1 \\ 0 \end{pmatrix} \quad (17)$$

- Impermeability tensor in the presence of E-field

$$\eta(\mathbf{E}) = \eta(\mathbf{0}) + \mathbf{rE} = \begin{pmatrix} 1/n_1^2 + r_{13}E_3 & 0 & r_{51}E_1 \\ 0 & 1/n_2^2 + r_{23}E_3 & r_{42}E_2 \\ r_{51}E_1 & r_{42}E_2 & 1/n_3^2 + r_{33}E_3 \end{pmatrix} \quad (18)$$

$$\eta(\mathbf{0}) = \text{diag}(1/n_1^2, 1/n_2^2, 1/n_3^2) \quad (19)$$

90°-domain PPKN Simulation

- **Propagating in the $X(b)$ -direction with an applied field $E_y(E_1)$**
for $+c$ and $-c$ domain of PPKN,

$$\eta_t(\mathbf{E}) = \begin{pmatrix} 1/n_1^2 \pm r_{13}E_3 & \pm r_{51}E_1 \\ \pm r_{51}E_1 & 1/n_3^2 \pm r_{33}E_3 \end{pmatrix} = \begin{pmatrix} 1/n_1^2 & \pm r_{51}E_1 \\ \pm r_{51}E_1 & 1/n_3^2 \end{pmatrix} \quad (20)$$

The rocking angle is defined as:

$$\tan 2\theta = \frac{2\eta_{t,12}}{\eta_{t,11} - \eta_{t,22}} = \frac{\pm 2r_{51}E_1}{(1/n_1^2) - (1/n_3^2)}$$

For small angle θ ,

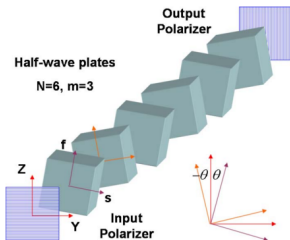
$$\theta \approx \frac{r_{51}E_1}{(1/n_1^2) - (1/n_3^2)} \quad (21)$$

Second order tensor - Impermeability tensor

Slicing the direction of $\mathbf{k} \cdot \mathbf{D} = 0$ on the index ellipsoid : transverse impermeability tensor

The information of transverse impermeability tensor :

polarization vector of normal modes (eigenvector of transverse impermeability tensor) with their refractive index.



Impermeability tensor of 90°-domain PPKN

Rotating Impermeability tensor of an new optic axis:

$$\eta'_{ij} = u_{ip} u_{jq} \eta_{pq} \quad (22)$$

where the tensor transformation $U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix}$ (23)

Impermeability tensor is crystallographically given by:

$$\eta = \begin{pmatrix} 1/n_1^2 & 0 & 0 \\ 0 & 1/n_2^2 & 0 \\ 0 & 0 & 1/n_3^2 \end{pmatrix} \quad (24)$$

Third order tensor - Pockels tensor

A third-order tensor r_{ijk} defined as the linear derivative of the impermeability tensor with respect to the electric field: $\Delta\eta_{ij} = r_{ijk}E_k$

$$\eta_{ij}(\mathbf{E}) = \eta_{ij}(0) + \left. \frac{\partial \eta_{ij}}{\partial E_k} \right|_{\mathbf{E}=0} E_k + \frac{1}{2} \left. \frac{\partial^2 \eta_{ij}}{\partial E_k \partial E_\ell} \right|_{\mathbf{E}=0} E_k E_\ell + \dots \quad (25)$$

All polar tensors of uneven rank vanish with the existence of an inversion center.

In our case, mm2 group is noncentrosymmetric.

Alive tensor components : $r_{113}, r_{223}, r_{333}, r_{131}$ and r_{232} .

Pockels tensor of 90°-domain PPKN

+c domain : $r_{113}, r_{223}, r_{333}, r_{131}$ and r_{232} .

Setting Pockels tensor of an arbitrary domain:

$$r'_{ijk} = u_{ip}u_{jq}u_{kr}r_{pqr} \quad (26)$$

where the tensor transformation $U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ u_{21} & u_{22} & u_{23} \\ u_{31} & u_{32} & u_{33} \end{pmatrix}$ (27)

New optic axis for Simulation (Lab) Setting

• 90°-domain PPKN structure

+c domain : Rotation by $+45^\circ$ in the $x'z'$ plane.

+a domain : Rotation by -45° in the $x'z'$ plane.

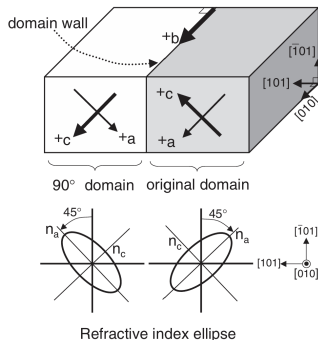


Fig. 2. Crystallographic and optical configurations between original domain and 90°-domain.

$$U_y(\theta) = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix} \quad (28)$$

Impermeability tensor of 90°-domain PPKN

- Unperturbed Impermeability tensor (in the absence of an electric field)

$\mathbf{U}_y(+\frac{\pi}{4})$ for $+c$ domain,

$$\eta'^{(+c)} = \begin{pmatrix} \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) & 0 & \frac{1}{2} \left(\frac{1}{n_3^2} - \frac{1}{n_1^2} \right) \\ 0 & 1/n_2^2 & 0 \\ \frac{1}{2} \left(\frac{1}{n_3^2} - \frac{1}{n_1^2} \right) & 0 & \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) \end{pmatrix} \quad (29)$$

$\mathbf{U}_y(-\frac{\pi}{4})$ for $+a$ domain

$$\eta'^{(+a)} = \begin{pmatrix} \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) & 0 & -\frac{1}{2} \left(\frac{1}{n_3^2} - \frac{1}{n_1^2} \right) \\ 0 & 1/n_2^2 & 0 \\ -\frac{1}{2} \left(\frac{1}{n_3^2} - \frac{1}{n_1^2} \right) & 0 & \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) \end{pmatrix} \quad (30)$$

Pockels tensor of 90°-domain PPKN

Rotation by φ for the Pockels tensor in new optic axis (r_{mk}),

$$r'_{mk}(\varphi)E_k = \begin{pmatrix} s(r_{13}c^2 + r_{33}s^2 + 2r_{51}c^2) & 0 & c(r_{13}c^2 + r_{33}s^2 - 2r_{51}s^2) \\ r_{23}s & 0 & r_{23}c \\ s(r_{13}s^2 + r_{33}c^2 - 2r_{51}c^2) & 0 & c(r_{13}s^2 + r_{33}c^2 + 2r_{51}s^2) \\ 0 & r_{42}c & 0 \\ c[(-r_{13} + r_{33} - 2r_{51})s^2 + r_{51}] & 0 & s[(-r_{13} + r_{33} - 2r_{51})c^2 + r_{51}] \\ 0 & r_{42}s & 0 \end{pmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} \quad (31)$$

Pockels tensor of 90°-domain PPKN

- With an applied field $E_y(E_2)$

$$r'_{mk}(\varphi)E_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ r_{42}E_2 \cos \varphi \\ 0 \\ r_{42}E_2 \sin \varphi \end{pmatrix} \quad (32)$$

Transverse Impermeability tensor in the presence of an electric field E_y

$$\eta'_t = \begin{pmatrix} 1/n_2^2 & r_{42}E_2 \cos \varphi \\ r_{42}E_2 \cos \varphi & \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) \end{pmatrix} \quad (33)$$

Pockels tensor of 90°-domain PPKN

- With an applied field $E_z(E_3)$

$$r'_{mk}(\varphi = \pm\pi/4) E_k = \begin{pmatrix} \frac{1}{\sqrt{2}} \left(\frac{1}{2}(r_{13} + r_{33}) - r_{51} \right) E \\ \frac{1}{\sqrt{2}} r_{23} E \\ \frac{1}{\sqrt{2}} \left(\frac{1}{2}(r_{13} + r_{33}) + r_{51} \right) E \\ 0 \\ \pm \frac{1}{\sqrt{2}} \left[\frac{1}{2}(-r_{13} + r_{33} - 2r_{51}) + r_{51} \right] E \\ 0 \end{pmatrix} \quad (34)$$

Transverse Impermeability tensor in the presence of an electric field E_y

$$\eta'_t = \begin{pmatrix} 1/n_2^2 + \frac{1}{\sqrt{2}} r_{23} E & 0 \\ 0 & \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) + \frac{1}{\sqrt{2}} \left(\frac{1}{2}(r_{13} + r_{33}) + r_{51} \right) E \end{pmatrix} \quad (35)$$

Comments on 90°-domain PPKN EO Šolc Filter

- **Case study**

Transverse Impermeability tensor in the presence of an electric field E_y

$$\eta'_t = \begin{pmatrix} 1/n_2^2 & r_{42}E_2 \cos \varphi \\ r_{42}E_2 \cos \varphi & \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) \end{pmatrix} \quad (36)$$

Two domains are rotated together in same axis.

Transverse Impermeability tensor in the presence of an electric field E_y

$$\eta'_t = \begin{pmatrix} 1/n_2^2 + \frac{1}{\sqrt{2}}r_{23}E & 0 \\ 0 & \frac{1}{2} \left(\frac{1}{n_1^2} + \frac{1}{n_3^2} \right) + \frac{1}{\sqrt{2}} \left(\frac{1}{2}(r_{13} + r_{33}) + r_{51} \right) E \end{pmatrix} \quad (37)$$

Two domains are not rotated.

Comments on 180°-domain PPKN EO Šolc Filter

- 180°-domain PPKN

Can neglect PVE effect for any incident polarized light.

Same equation for rocking angle (PPKN and PPLN)

$$\theta \approx \frac{r_{51}E}{(1/n_1^2) - (1/n_3^2)}, \quad \theta \approx \frac{r_{51}E}{(1/n_e^2) - (1/n_o^2)} \quad (38)$$

Unclamped Pockels coefficient r_{51} of PPKN : (120 ± 10) pm/V

Unclamped Pockels coefficient r_{51} of PPLN : 32.6 pm/V

In the C-band (1550nm) with an arbitrary amplitude of E-field,

θ of PPLN : $(1.27 \times 10^{-7} E)^\circ$

θ of PPKN : $(3.56 \times 10^{-7} E)^\circ$

Simulation on PPKN EO Šolc filter

- **180°-domain PPKN**

Solc design @ $T = 24.00\text{ }^{\circ}\text{C}$, target $\lambda_0 = 1550.00\text{ nm}$

Crystal size = $28.00\text{ mm} \times 5.00\text{ mm} \times 0.50\text{ mm}$

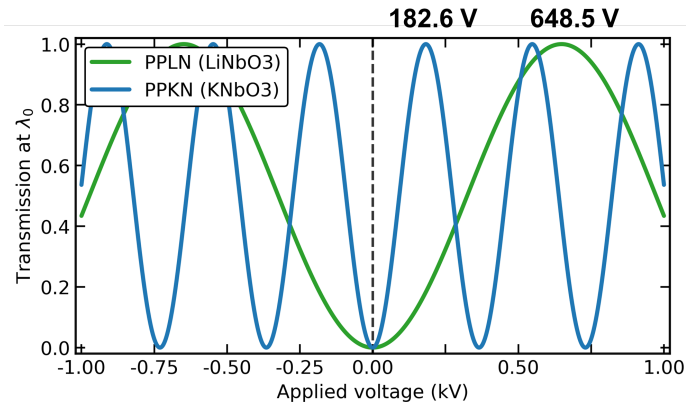
Table: Comparison of periodically poled Solc-filter parameters at 1550 nm.

Crystal	N (domains)	Λ (μm)	λ_0 (nm)
PPLN (LiNbO_3)	2729	20.520 337	1550.119
PPKN (KNbO_3)	3464	16.166 282	1550.118

Simulation on PPKN EO Šolc filter

- 180°-domain PPKN

$$V_{\pi,KN} \approx 3V_{\pi,LN}$$



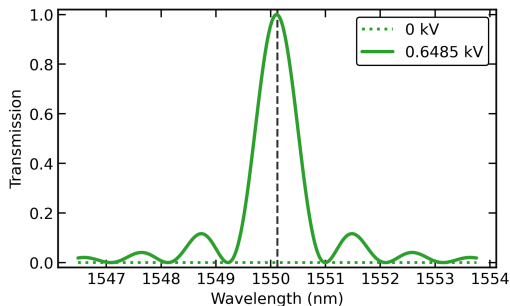
Simulation on PPKN EO Šolc filter

- 180° -domain PPKN

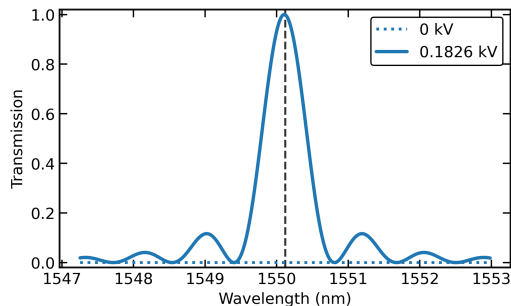
$$V_{\pi,KN} \approx 3V_{\pi,LN}$$

Green line : PPLN, Blue line : PPKN

FWHM=0.9088 nm

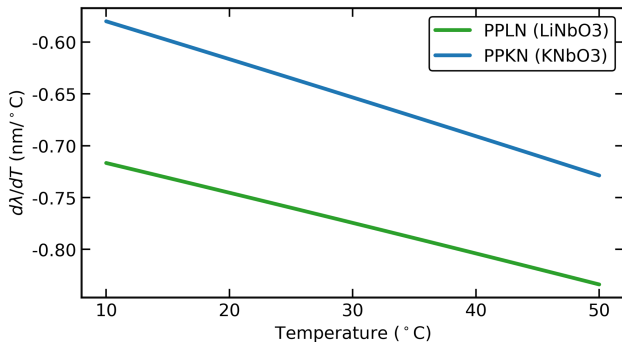


FWHM=0.716 nm



Simulation on PPKN EO Šolc filter

- Temperature dependence of 180°-domain PPKN



Temperature (°C)	PPLN nm/°C	PPKN nm/°C
10	-0.716 660	-0.579 997
24	-0.756 933	-0.631 209
50	-0.833 817	-0.728 711

Simulation on PPKN EO Šolc filter

- **180°-domain PPKN**

The temperature moves central wavelength.

